

CONTACT	Orlando, FL mdelias.hossain@ucf.edu	Google Scholar Website GitHub LinkedIn
RESEARCH SUMMARY	PhD researcher building trustworthy and safe large language models and AI agents. I develop and evaluate methods for LLM alignment under corrupted preference supervision (RLHF/DPO), calibrated uncertainty for foundation models, runtime safety for tool-using agents, and inference-time control of frozen policies, and I ship the systems, benchmarks, and open-source code behind them. Lead author with work accepted at ICLR and under review at NeurIPS and ACL. 937 Google Scholar citations h-index 12 14+ open-source repositories 4 public benchmarks Areas: Foundation Models LLM Alignment (RLHF, DPO) AI Agents & Agent Safety Uncertainty Quantification & Calibration Inference-Time Control Offline RL LLM Evaluation & Benchmarking	
EDUCATION	Ph.D., Industrial Engineering (CS concentration), University of Central Florida Aug 2025 – Present Advisor: Dr. Niloofar Yousefi. GPA 3.87/4.00. Passed PhD qualifying exam; advancing to candidacy Fall 2026. Research: trustworthy AI for LLMs (alignment, uncertainty, agent safety). Expected graduation Spring 2028. M.S., Computer Science (Research), Mississippi State University Jan 2024 – Jul 2025 GPA 3.88/4.00. Advisors: Dr. Andy D. Perkins, Dr. Shahram Rahimi. B.S., Software Engineering, Daffodil International University 2016 – 2020	
EXPERIENCE	Graduate Research Assistant, UCF Complex Adaptive Systems Lab Aug 2025 – Present - Lead an independent research program on trustworthy LLMs and agent safety, advised by Dr. Niloofar Yousefi and collaborating with Prof. Ser-Nam Lim (formerly Meta AI / FAIR); first-author work accepted at ICLR 2026 (Trustworthy AI Workshop) and under review at NeurIPS and ACL. - Designed and implemented NEXUS, a runtime safety monitor for tool-using LLM agents, raising intervention accuracy +27.3 points over rule-only baselines; released the code and 4 public benchmarks (74,590+ examples). Graduate Research Assistant, Mississippi State University May 2025 – Jul 2025 - Built and deployed BarkPlug V.2, a production RAG chatbot; engineered FAISS retrieval and evaluated with RAGAS and the System Usability Scale. - Developed a contrastive-learning framework for anomaly detection over large-scale cybersecurity event logs. Software Engineer, REVE Systems, Dhaka, Bangladesh Dec 2022 – Nov 2023 - Trained and deployed production speech-recognition models (DeepSpeech, Wav2Vec 2.0): 20% lower Word Error Rate and 15% faster inference; shipped a voice extension for national e-governance services. Senior AI/ML Researcher (Remote), Time Research & Innovation, Portsmouth, UK Oct 2020 – Nov 2022 - Designed and trained deep-learning architectures for COVID-19 diagnosis from X-ray/CT imagery; co-built a UK telemedicine platform.	
SELECTED PUBLICATIONS	<i>Published or accepted, first author unless noted. Full list on Google Scholar.</i> - Introduced multi-agent verification of preference supervision for RLHF/DPO robust to corrupted feedback. <i>ICLR 2026 Trustworthy AI Workshop</i> (accepted). - Synthesized failure modes, evaluation protocols, and safety mechanisms for collaborative multi-agent LLM systems. <i>IEEE Trans. on Systems, Man, and Cybernetics: Systems</i>, 2026. [link]	

- Built **BIOGEN**, an evidence-grounded multi-agent reasoning framework for transcriptomic interpretation. *Frontiers in Bioinformatics*, 2026. [link]
- Designed **R-GAT**, a relational graph-attention model for document classification, and released the CancerAbstracts benchmark. *Scientific Reports* (Nature Portfolio), 2026. [link]
- First-author systematic review of **NLP in electronic health records (513 citations)**. *Computers in Biology and Medicine*, 2023. [link]

FIRST-AUTHOR MANUSCRIPTS UNDER REVIEW

Submitted, not yet accepted.

- Built **NEXUS**: deterministic rules, argument-level inspection, and a calibrated risk scorer route agent plans to graded interventions. *ACL ARR 2026*. [Code]
- Developed **UAT-LITE**, inference-time uncertainty-aware attention that calibrates pretrained transformers without retraining. *ACL ARR 2026*. [arXiv]
- Proved a unified closed-form view of **post-training steering in offline RL** for policies that cannot be retrained. *NeurIPS 2026 (Main Track)*.
- Developed **MedBayes-Lite**, Bayesian uncertainty quantification for safe clinical decision support. *Under review*, 2026. [arXiv]

SELECTED PROJECTS, BENCHMARKS & OPEN SOURCE

Released reproducible code, datasets, and evaluation pipelines for all major research projects. huggingface.co/EliasHossain | github.com/eliashossain001

- **CorrFilter** (with **Prof. Ser-Nam Lim**, formerly Meta FAIR): measured that LLM-judge banks used for alignment-data filtering fail dependently, so consensus filtering degrades as judges correlate (effective independent judges stay near 5 of 16); built a calibration-based adaptive filter that reduces false retention by up to **6 points**.
- **Released 4 public benchmarks (74,590+ examples)** on Hugging Face Hub: schema-constrained biomedical extraction (**nanobubbleeval**, 51.6K), biomedical contradiction NLI (**BioDivergence**, 11.9K), knowledge-conflict temporal QA (**ptc-benchmark**, 9.25K), and cancer-abstract classification (**CancerAbstracts**).
- **Implemented inference-time steering of frozen policies**: **PoE-Composition** (product-of-experts refinement for offline RL) and **Temporal Attractor Steering** (detects and corrects parametric temporal conflicts without retraining).
- **Built a Qwen-style 0.6B transformer from scratch in PyTorch (qwen-scratch-0.6B)**: tokenizer, attention, and training loop end to end.
- **Built LLM training pipelines**: **LoRA/QLoRA (PEFT) fine-tuning** on multi-GPU and **continual pretraining** for domain adaptation without catastrophic forgetting.

TECHNICAL SKILLS

- **LLMs & Alignment**: fine-tuning and alignment (RLHF, DPO, SFT); PEFT (LoRA/QLoRA); AI agents and tool use; inference-time steering/control of frozen models; continual pretraining; offline RL; multimodal modeling.
- **Trustworthy AI & Evaluation**: uncertainty quantification, calibration, Bayesian learning; LLM evaluation and benchmarking; runtime safety and alignment; adversarial robustness (prompt injection, data poisoning); graph neural networks.
- **Systems & Tools**: Python, R, SQL, Bash; PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn; multi-GPU / distributed training; FAISS, Chroma, RAG / retrieval; Apache Spark; Docker, Git, Azure, GCP.

RECOGNITION & SERVICE

- **Talks**: ICLR 2026 Trustworthy AI Workshop; MoML, MIT Jameel Clinic (2025); Amazon Trusted AI Symposium poster (2026).
- **Awards**: UCF Graduate Presentation Fellowship (2025); Finalist, Dr. Pradeep P. Thevannoor Innovation Awards (2019).
- **Service & Teaching**: Reviewer for ICLR Trustworthy AI Workshop (2026) and 7 journals (Nature Portfolio, Elsevier, PLOS, SAGE, Springer); Graduate TA in ML/AI.